

* Regularization Technique

What is it?

Regularization adds a penalty or constraint to the learning process to discourage overly complex models that fit noise in the training data.

Goal:- Improve generalization to unseen data

@codingdidi

Overfitting
(high variance)



Just Right
(good generalization)



Underfitting
(high bias)



Regularization pushes model complexity to the "just right" zone.

Why used?

- ✓ Prevent Overfitting.
- ✓ Improve generalization to new data.
- ✓ Handles high-dimensional data.
- ✓ Make model robust to noise.
- ✓ Encourage simpler, more interpretable models.

✓ Stabilize training and improve performance on validation / test sets.

We don't want a model that only explains the training data well, but one that explains the true underlying pattern.

When Used?

(avoiding)

- * When training error is low but validation / test error is high $\rightarrow$ overfitting.
- * When model is complex (many problems, many parameters, deep networks, polynomial etc.)
- * When dataset is small or noisy.
- * When there are multiple features (high dimensionality).
- * When you want simpler, more robust models.
- * Typically tuned using validation set / cross-validation.

Rule of Thumb :- Try a simple model first. Add regularisation when overfitting appears.